

FDS: GetOffMyTrain — Project Report

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1. Introduction and Related work

Board games provide a structured yet visually complex environment that is well suited for computer vision and data science applications. Ticket to Ride, in particular, poses non-trivial challenges due to the presence of multiple players, colored pieces, overlapping routes, and strong variability in real-world acquisition conditions such as camera angle and lighting.

The goal of this project is to automatically reconstruct the state of a Ticket to Ride game from a single photograph of the board. This task is motivated by potential applications such as automatic score tracking, game analysis, and augmented reality support. From a data science perspective, the problem requires combining image understanding with structured data extraction to produce a formal representation of the game state.

Recent advances in deep learning have significantly improved object detection in complex scenes. Models such as YOLO have shown strong performance in real-time detection tasks and are widely adopted due to their accuracy and efficiency [1]. Prior work has applied computer vision techniques to board games such as chess or Go, often under controlled conditions and with simpler layouts [2]. However, fewer approaches address unconstrained images of complex board games like Ticket to Ride, where both geometric normalization and fine-grained object recognition are required.

More recently, large multimodal models have demonstrated zero-shot image understanding capabilities. While promising, our preliminary experiments indicate that such models struggle to produce accurate and consistent structured outputs for this task, motivating the use of a dedicated, task-specific pipeline.

In this work, we propose a two-stage approach that leverages deep learning for board alignment and train detection, producing a structured JSON representation of the game state suitable for downstream analysis.

2. Proposed method

The proposed system is designed as a two-stage pipeline that reconstructs the state of a Ticket to Ride game from a single image of the board. The core idea is to first normalize the board geometry and then perform train and route detection on a standardized representation. This separation allows the system to be modular, robust to acquisition variability, and easier to extend.

Deep learning is used in both stages through two independent YOLOv8 models, each specialized for a specific task.

2.1. Stage 1: Board Detection and Perspective Normalization

In the first stage, a YOLOv8 model is trained to detect the four corner points of the game board. This step is necessary because images are captured under unconstrained conditions, often with perspective distortion due to camera angle and distance.

Once the four corners are identified, a homography transformation is computed to map the original image onto a canonical board template. This geometric normalization produces a consistent, aligned view of the board, reducing spatial variability and simplifying subsequent detection tasks.

2.2. Stage 2: Train Detection and Route Assignment

In the second stage, a second YOLOv8 model operates on the normalized board image to detect trains and classify their colors, corresponding to the different players. For each detected train, the bounding box center is extracted.

Route definitions are provided through a predefined JSON file containing the geometric representation of each route on the board. By mapping detected train positions to these route regions, the system assigns trains to specific routes. Finally, all detected information is aggregated into a structured JSON representation describing the current game state, including the partial scores of the players.

3. Dataset and Benchmark

3.1. Dataset

The dataset used in this project is a public dataset of Ticket to Ride USA board images available on Kaggle [3]. It consists of 40 images of the game board captured under different conditions, including varying camera angles, distances, and lighting setups, reflecting a realistic and unconstrained acquisition scenario. Each image was manually annotated to support both stages of the pipeline. Annotations include the four corner points of the board and the bounding boxes (or masks) of the trains, along with their corresponding player colors. Annotation and data augmentation were managed using Roboflow [4], with an additional manual refinement step to improve annotation quality. Due to the limited dataset size, it was not possible to create a separate validation set. The data were split into training and test sets, with different proportions depending on the task.

3.2. Train/Test Split

For the corner detection task, the dataset was split evenly into training and test sets (50/50), prioritizing generalization across different perspectives. For train and color detection, a 75/25 train/test split was adopted to provide sufficient training samples while retaining a meaningful test set.

3.3. Benchmark

As a qualitative benchmark, we evaluated a zero-shot approach using a large multimodal model (Gemini [5]). The model was provided with a board image and a JSON description of the routes and asked to infer the game state without task-specific training. While this approach demonstrates the potential of general-purpose models, it produced frequent errors in route occupancy and color assignment, leading to incorrect score estimation. This comparison highlights the advantage of a dedicated, task-specific pipeline over zero-shot solutions for structured game state reconstruction.

4. Experimental results

4.1. Evaluation metrics

We evaluated the performance of the two YOLOv8 models using standard object detection metrics, including Precision, Recall, and mean Average Precision (mAP) at different IoU thresholds. Metrics were computed separately for the corner detection task and for the train and color detection task, reflecting the modular design of the pipeline.

4.2. Corner Detection Results

The corner detection model achieved strong performance, with an mAP@50 of 0.92 and an mAP@50–95 of 0.85.

Precision and Recall were both high (0.88 and 0.90, respectively), indicating reliable detection of board corners across different acquisition conditions. These results confirm that the model effectively learns the geometric structure of the board, which is crucial for accurate perspective normalization. Errors in this stage were rare and typically associated with extreme viewpoints or partial occlusions.

4.3. Train and Color Detection Results

The train detection model obtained an mAP@50 of 0.78 and an mAP@50–95 of 0.65, with Precision of 0.81 and Recall of 0.74. While overall performance is satisfactory given the limited dataset size, this stage proved more challenging due to the small size of trains, visual similarity between colors, and overlapping routes. In particular, misclassifications often involved confusion between visually similar colors or closely spaced parallel routes.

4.4. Results Analysis

Despite the relatively small dataset, the results demonstrate that the proposed approach is effective and robust. Corner detection performs particularly well, validating the decision to separate geometric normalization from object detection. Train detection errors, while present, highlight realistic limitations of the data and suggest that additional training samples and fine-tuning could further improve performance. Overall, the system produces reliable structured game states suitable for downstream analysis.

5. Conclusions

In this work, we presented a fully automated pipeline for reconstructing the state of a Ticket to Ride game from a single photograph of the board. By separating geometric normalization from object and color detection, the proposed approach achieves robustness to variations in camera angle, lighting conditions, and game configurations.

Experimental results show that corner detection is highly reliable, confirming the effectiveness of using a dedicated deep learning model for board alignment. Train and color detection, while more challenging due to visual ambiguity and limited data, produces sufficiently accurate results to reconstruct a structured game state. Importantly, the comparison with a zero-shot multimodal model highlights that general-purpose approaches struggle to provide consistent and precise structured outputs for this task.

Overall, this project demonstrates that combining deep learning with domain-specific constraints is essential for reliable game state extraction in complex board games. The proposed system is modular, operational, and provides a solid foundation for future extensions such as dataset expansion, model fine-tuning, augmented reality integration, and AI-based gameplay analysis. [Github repository](#)

References

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